

Context Compression Techniques for Legal Document Analysis

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Abstract

Legal application are bound to the handling of large scale documnt analysis when starting a case. The frontier models that powers the majority of LLM applications are reporting on increasing maximum sequece length / context windows. But the

Chapter 1

Introduction

The legal industry is increasingly adopting large language models (LLMs) to assist lawyers in navigating the complexity of modern litigation. Legal cases are inherently dynamic: they evolve over time as new documents, correspondence, evidence, and procedural developments continuously emerge. What was once a central claim may become peripheral; a newly submitted document may reframe the entire dispute. This progression means that not only does the volume of material grow, but the *relevance* and *weight* of existing information shifts as the case develops.

A fundamental limitation of current LLM-based systems is that they are stateless. Without explicit mechanisms for maintaining case state, models must re-read the full document history for every query. More critically, even models that handle large contexts tend to perform well in early stages but *drift* over extended, multi-turn interactions — gradually losing track of what is current, what has been superseded, and what actually matters. In legal case management, this drift is not merely an inconvenience; it can lead to factually incorrect summaries, missed deadlines, or misrepresentation of parties and claims.

This thesis addresses this problem through two complementary mechanisms: *compression* of case-relevant information into a structured representation, and *state maintenance* — the continuous, structured tracking of case essentials as the project evolves. The vehicle for this is a structured *FactSheet*: a living, machine-readable summary of the case that is initialized from initial input and updated incrementally as new documents and information arrive.

The research investigates whether maintaining an explicit, structured state representation improves an LLM agent’s ability to preserve contextual accuracy over the full lifecycle of a legal case — particularly as conversation length grows and the document landscape changes. The system is implemented as a LangGraph-based conversational agent, incorporating multi-phase document processing pipelines for both project initialization and incremental updates.

The contributions of this thesis are: (1) the design and implementation of initialization and update pipelines for structured legal case state management, (2) an evaluation framework comparing structured state agents against RAG-only and baseline approaches, and (3) an analysis of accuracy degradation, compression efficiency, and state consistency across extended multi-turn conversations.

The remaining chapters are organized as follows: Chapter 2 presents background and related work on long-context LLMs, context compression, and retrieval-augmented generation. Chapter 3 describes the methodology, system architecture, and evaluation design. Chapter 4 presents experimental results. Chapter 5 discusses findings and implications, and Chapter 6 concludes with a summary and directions for future work.

Chapter 2

Theory

This chapter presents the theoretical foundation for the approaches explored in this thesis. We examine the core challenges of applying large language models to long-context, multi-turn legal document analysis, and survey the relevant literature on retrieval, compression, and agentic context management strategies.

2.1 Large Language Models and Context Windows

Modern large language models (LLMs) are built on the transformer architecture [VSP⁺23], which processes input as a sequence of tokens bounded by a fixed context window. While recent models advertise longer context windows, empirical benchmarks reveal a significant gap between nominal and effective context utilization. introduce RULER, a benchmark designed to measure the *real* usable context size of long-context LLMs, demonstrating that performance degrades substantially before the advertised limit is reached [HSK⁺24]. This has direct implications for legal use cases, where case files can span hundreds of pages across dozens of documents.

2.2 The Lost in the Middle Problem

Even within the effective context window, LLMs do not attend uniformly to all input. Liu et al. show that model performance is highest when relevant information appears at the beginning or end of the context, and degrades for information placed in the middle — a phenomenon they term *lost in the middle* [LLH⁺23]. This positional bias has critical consequences for legal document analysis, where key facts may be embedded deep within lengthy attachments. It motivates the need for strategies that either restructure context placement or reduce reliance on verbatim document injection altogether.

2.3 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) addresses context limitations by retrieving only the most relevant document chunks at query time, rather than loading entire documents into the context window [LPP⁺21]. A standard RAG pipeline consists of offline chunking and embedding of documents into a vector store, followed by semantic retrieval at inference time. While effective for single-turn question answering, naive RAG struggles in multi-turn conversational settings where relevance depends on accumulated context across turns — a central challenge in this thesis.

RECOMP [XSC23] extends the RAG paradigm with both extractive and abstractive compressors that filter and condense retrieved passages before they are passed to the generator, reducing noise and token consumption while preserving relevant information.

2.4 Prompt Compression

An orthogonal approach to managing long contexts is prompt compression: reducing the token footprint of the input itself rather than selectively retrieving. LLMLingua [JWL⁺23] proposes a coarse-to-

fine compression method guided by a smaller language model, identifying and removing tokens with low information content. LongLLMLingua [JWL⁺24] extends this to long-document scenarios, incorporating question-aware compression that prioritizes tokens contextually relevant to the user query. Both methods are relevant as baseline comparisons for the context management strategies implemented in this thesis.

2.5 Graph-Based and Multi-Hop Retrieval

Standard vector similarity retrieval operates on isolated chunks and does not capture relational structure between entities. In legal cases, facts, parties, events, and claims are deeply interconnected, making relational retrieval particularly valuable. GraphRAG [?] proposes building a knowledge graph over the document corpus, enabling community-level summarization and structured traversal of entity relationships. HopRAG [LWC⁺25] further extends this with multi-hop reasoning, constructing logical chains across documents to answer complex queries that require synthesizing information from multiple sources. These approaches are particularly relevant to the factsheet extraction pipeline implemented in this thesis, where structured relationships between parties, events, and claims must be maintained across turns.

2.6 Infinite and Extended Context Architectures

An alternative to prompt-level strategies is modifying the underlying model architecture to support longer effective memory. Munkhdalai et al. propose Infini-Attention, which introduces compressive memory into the transformer attention mechanism to enable theoretically unbounded context without quadratic scaling [MFG24]. While this represents a promising long-term direction, it requires architectural changes to the model and is not applicable in API-based deployment settings, which is the constraint under which this thesis operates.

2.7 Agentic Context Management

Beyond static retrieval and compression, agentic systems can actively manage context across multi-turn conversations. Rolling summarization — periodically condensing conversation history into a compact summary — is a practical strategy for preventing context overflow in long sessions [Mic24]. Stateful agent frameworks such as LangGraph enable persistent state management through checkpointing, allowing the agent to maintain structured representations (e.g., factsheets) that accumulate case knowledge across turns without requiring the full document history to remain in context.

Taken together, the strategies surveyed in this chapter — effective context benchmarking, positional bias awareness, RAG with compression, graph-based retrieval, and agentic summarization — form the theoretical basis for the system design and experimental comparisons presented in the following chapters.

Chapter 3

Method

3.1 System Architecture

The system is designed as a modular, client-server application consisting of four main packages organized as a `uv` workspace: `agent`, `ui`, `evals`, and `data-viewer`. The architecture follows a separation of concerns, where the backend handles all agent logic, data persistence, and document processing, while the frontend is responsible for user interaction and result presentation.

3.1.1 Overview

The backend is implemented as a **FastAPI** application (`agent/main.py`) that exposes a RESTful API with Server-Sent Events (SSE) for real-time response streaming. The frontend is a **Streamlit** application that communicates with the backend over HTTP, consuming streamed responses. Authentication is handled via **Supabase Auth** using JWT tokens, which are validated on each request through a `HTTPBearer` dependency in FastAPI.

3.1.2 Agent Layer

The core agent is implemented using **LangGraph's StateGraph**, a directed graph framework for building stateful LLM applications. The agent graph consists of two nodes:

- `call_llm` — Constructs the full input payload, including the system prompt, factsheet context, conversation history, and RAG-retrieved document chunks, then invokes the LLM.
- `call_tool` — Executes any tool calls returned by the LLM.

A conditional edge routes control back to `call_tool` if the LLM emits tool calls, or terminates the graph otherwise. To manage long conversations within the model's context window, a rolling summarization strategy is applied every eight messages using a dedicated **Summarizer** module.

Conversation state is persisted using LangGraph's **AsyncPostgresSaver** checkpoint, backed by **Supabase PostgreSQL**, enabling session resumption and multi-turn dialogue across requests.

3.1.3 Tool Layer

The agent has access to a set of callable tools defined in `tools.py`. The full tool set (**TOOLS**) used in the primary agent configuration includes:

- `tavily_search` — Web search via the Tavily API.
- `read_attachment` — Fetches and parses a document from Supabase Storage.
- `query_project_attachments` — Performs semantic retrieval (RAG) against project-specific documents indexed in BigQuery Vector Store.
- `read_laws` — Semantic retrieval against a corpus of Norwegian legislation.

- `update_project`, `clean_element`, `create_project` — Tools for managing the factsheet state.

Two ablated tool configurations are also defined for evaluation purposes: **BASELINE_TOOLS** (web search and law retrieval only) and **BASELINE_RAG_TOOLS** (web search, law retrieval, and project document RAG, but without file storage access).

3.1.4 Data and Storage Layer

Document persistence is handled through two complementary mechanisms. Raw files (PDF, DOCX, PPTX, EML) are stored in **Supabase Storage**, where they can be retrieved on demand by the `read_attachment` tool. Parsed and chunked document content is embedded using **Google’s gemini-embedding-001 model** and stored in **BigQuery Vector Store** for semantic retrieval. Structured project data — including the **FactSheet** and session metadata — is stored in Supabase PostgreSQL via the **SupabaseManager** CRUD interface.

3.1.5 Document Processing Pipeline

Upon project initialization (`POST /init-project`), a multi-phase asynchronous pipeline is triggered:

1. **Phase 1:** Documents are parsed using **DocumentProcessor** (supporting PDF with OCR, DOCX, PPTX, and EML formats), stored in file storage, and indexed in the vector store.
2. **Phase 2:** Documents and emails are analyzed in parallel to extract structured entities — events, claims, damages, deadlines, and attachment metadata.
3. **Phase 3:** Factual facts and governing law are synthesized across all documents.

Status events are streamed back to the frontend via SSE throughout all phases.

3.1.6 LLM Configuration

The system supports multiple LLM providers through LangChain’s `init_chat_model` interface. **Google Gemini** serves as the primary provider, with **OpenAI GPT** available as a secondary option. The active model is configurable per session request, allowing for flexible experimentation and evaluation across providers.

3.1.7 Tech Stack Summary

Layer	Technology
Language	Python 3.13
Agent framework	LangGraph (StateGraph)
LLM orchestration	LangChain (<code>init_chat_model</code>)
LLM providers	Google Gemini (primary), OpenAI GPT (secondary)
Backend API	FastAPI + Uvicorn
Streaming	Server-Sent Events (SSE)
Frontend	Streamlit
Authentication	Supabase Auth (JWT)
Database	Supabase (PostgreSQL)
Vector store	BigQuery Vector Store (persistent)
File storage	Supabase Storage
Conversation checkpointer	LangGraph <code>AsyncPostgresSaver</code>
Document parsing	PyPDF2, ocrmypdf, python-docx, python-pptx
Embeddings	Google <code>gemini-embedding-001</code>
Web search	Tavily

Table 3.1: System tech stack.

3.2 Design Choices

The system architecture was guided by several pragmatic design choices, prioritizing development speed, flexibility, and cost-efficiency given the constraints of a master’s thesis. By separating the descriptive architecture from the rationale, the core motivations for the selected technologies are outlined below.

Agent Layer and Orchestration

The core agent is implemented using **LangGraph** [Lan24]. This framework was chosen primarily for its fast build time, which is essential given the limited time available for this thesis. It provides a clean **StateGraph** abstraction that is highly expandable for future extensions, such as adding dedicated reasoning nodes or document re-ranking stages. Furthermore, LangGraph is model-agnostic, providing built-in support for multiple LLM providers. This flexibility is crucial, allowing the system to easily switch between models or add new ones as desired for evaluation and experimentation.

Database and Storage Architecture

The system’s persistence layer separates structured relational data, unstructured document storage, and semantic vector retrieval to optimize for specific access patterns and constraints.

Relational Storage (PostgreSQL via Supabase): PostgreSQL was selected for its flexibility and highly efficient reads across tables. The structured nature of the factsheet state requires strict enforcement of relationships through primary and foreign keys to maintain data integrity and orderly matching on IDs [Sup20]. An initial attempt was made using a NoSQL database (Firestore); while easy and fast to set up, the lack of an enforced schema led to a loss of overview, with state inconsistencies and errors accumulating over time. Additionally, performing complex reads and combining data across different entities proved expensive and inefficient in NoSQL.

Supabase was chosen as the PostgreSQL provider because it is affordable, open-source, and seamlessly integrates other necessary products like Authentication. Having these features included out-of-the-box makes the development process straightforward and fast.

Database Schema

The database schema is designed to support the agent’s state management and document metadata, while being flexible enough to accommodate future extensions. There are three main groupings of tables:

Project related tables
Projects
Parties
Events
Deadlines
Attachments
Emails
Claims (legal specific)
Damages (legal specific)

Session related tables
Sessions
Session Events
Attachments

Tables for storing document metadata and relationships.

Agent checkpoints
Checkpoints
Writes
Blobs

Tables for the agent’s memory and conversation history.

These tables are combined to form the factsheet state, which is the agent’s internal representation of the case. The factsheet is updated and referenced throughout the agent’s operation, and it is crucial that it is stored in a structured and consistent manner.

File Storage (Supabase Storage): Raw documents (PDF, DOCX, EML, etc.) are stored in Supabase Storage. There were no specific architectural requirements for this other than cost-efficiency. Since Supabase was already selected for the PostgreSQL database, using its built-in, affordable object storage was a natural and convenient choice [Sup20].

Vector Store and Embeddings: Semantic retrieval relies on **Google’s embedding model** (`gemini-embedding-001`). This model was specifically chosen for its strong multilingual capabilities; smaller and cheaper open-source models often struggle with RAG matching when queries or documents are not strictly in English, which is a critical requirement for handling Norwegian legal texts.

For the vector database, a two-tier approach is used. **ChromaDB** is utilized as an in-memory store for fast, session-scoped retrieval. For persistent, project-scoped retrieval, **BigQuery Vector Search** is employed. BigQuery was chosen because it is affordable and operates on an on-demand pricing model. The primary drawback of BigQuery is its high latency, meaning a significant amount of time is spent indexing and saving to the vector store. However, this trade-off is worthwhile: the alternative would require provisioning an always-running vector database server, which introduces high continuous costs and additional operational setup.

FastAPI and Server-Sent Events

The backend is exposed as a standalone REST API using **FastAPI** [Ram18] to decouple it from the frontend. **Server-Sent Events (SSE)** are used for streaming responses because the agent pipeline is long-running and asynchronous. SSE provides a responsive user experience by pushing incremental status updates and partial results without the complexity of WebSockets, which are unnecessary for this unidirectional communication flow.

3.3 Evaluation Design

Describe the evaluation design, including the datasets used, metrics for evaluation, and the experimental setup.

3.3.1 Datasets

The evaluation framework relies on three custom-built datasets developed specifically for this thesis in collaboration with legal experts from the law firm BAHR AS. Each dataset is derived from a unique, real-world legal case processed through the Norwegian district court system. While the final verdicts and formal submissions from both parties were publicly available, the underlying case documents (e.g., correspondence, internal notes, and evidence) remained confidential.

To facilitate a robust evaluation of the RAG (Retrieval-Augmented Generation) system, we have reconstructed these cases by generating synthetic document sets. These sets include simulated emails, technical reports, agreements, and contracts, all meticulously crafted to mirror the factual matrix and legal complexities described in the official court records. This approach ensures that the agent is tested against a realistic representation of a legal case file while maintaining necessary privacy standards.

The datasets are organized to simulate the chronological lifecycle of a legal proceeding. Each case is structured into several *sessions*, each containing a sequence of *queries*. This structure is designed to evaluate two critical capabilities of the agent:

1. **Contextual Awareness:** The ability to maintain a coherent understanding of the case as it evolves through different stages of litigation.
2. **Analytical Accuracy:** Each query is paired with a "gold standard" expected output, defined by legal experts. These references serve as the ground truth for measuring correctness, relevancy, and completeness.

By following a chronological order, the sessions require the agent to handle increasing amounts of information and maintain consistency over time, reflecting the professional requirements of a legal practitioner managing an ongoing case.

3.3.2 Evaluation Environment

Evaluation Framework:

The evaluation will be conducted using **DeepEval** [Con23], a framework for evaluating LLM outputs against reference answers. DeepEval allows for automatic scoring based.

Agent Configurations:

There are three agent configurations to be evaluated:

- **Custom Agent:** The full agent configuration with access to all tools, including file storage and RAG capabilities.
- **Baseline Agent:** An ablated version of the agent with access to only web search and law retrieval tools, without file storage or RAG capabilities.
- **Baseline + RAG Agent:** An ablated version of the agent with access to web search, law retrieval, and project document RAG tools, but without file storage access.

Model Configuration:

Since the scope of the thesis is not to evaluate the models themselves, but rather the design and architecture of the agent, a selection of four models is used for comparison. These include both closed-source and open-source variants to ensure the agent’s performance is consistent across different provider architectures.

Table 3.2: Overview of Large Language Models used in evaluation.

Model Name	Type	Total Params	Active Params	Provider
Gemini 2.5 Pro	Closed	Unknown	Unknown	Google
Gemini 2.5 Flash	Closed	Unknown	Unknown	Google
Llama 4 Maverick [ba26]	Open (MoE)	400B	17B	Meta
Qwen 3.5 [YLY+25]	Open (MoE)	397B	17B	Alibaba
Nemotron-3 Nano [NB+25]	Open (MoE)	31.6B	3.2B	NVIDIA

The evaluation will be conducted across all datasets using consistent model-agent pairings. To account for the inherent stochastic nature of LLM outputs, each evaluation set (model + agent configuration) will be executed 5 times to ensure statistical significance in the results.

3.3.3 Metrics

The evaluation will focus on several key metrics, measuring both the quality of the agent’s responses and its efficiency in terms of resource usage. Metrics that are to be evaluated are:

- **Correctness, Relevancy, and Completeness:** These semantic metrics are evaluated using the DeepEval framework. The agent’s output is compared against a ”gold standard” reference created in collaboration with legal experts. Each metric is scored on a scale from 0 to 1.
- **Self-consistency:** The degree to which the agent’s output remains consistent across multiple queries and over time as the case evolves.

$$\text{Self-Consistency} = \frac{2}{n(n-1)} \sum_{i=1}^{n-1} \sum_{j=i+1}^n \cos(\mathbf{e}_i, \mathbf{e}_j) \quad (3.1)$$

where n is the number of generated outputs, $\mathbf{e}_i$ and $\mathbf{e}_j$ are the sentence embeddings of the i -th and j -th output, and the cosine similarity is defined as:

$$\cos(\mathbf{e}_i, \mathbf{e}_j) = \frac{\mathbf{e}_i \cdot \mathbf{e}_j}{\|\mathbf{e}_i\| \|\mathbf{e}_j\|} \quad (3.2)$$

- **BERTScore:** A metric that evaluates the similarity between the agent’s output and a reference text using BERT embeddings [ZKW+20].

- **Passrate:** The percentage of queries for which the agent’s output meets a predefined quality threshold.
- **Efficiency Metrics:** **Token Usage** (total tokens consumed) and **Inference Time** (latency) are recorded to assess the practical viability of the system.

Correctness, relevancy, Completeness are all evaluated against the expected out (constructed in collaboration with legal experts) using DeepEval. The framework rates each response on a scale from 0 to 1, with a certain threshold for what is considered a "pass" for the passrate metric.

BERTScore is calculated based on the expected output. Self-consistency is evaluated by comparing the agent’s responses across multiple queries and sessions to assess whether it maintains a coherent understanding of the case over time. Token usage and inference time are recorded during the agent’s operation.

3.3.4 One-sided Hypothesis Tests

For quality/performance metrics (correctness, relevancy, completeness, passrate, self-consistency, BERTScore):

$$H_0 : \mu_{\text{custom}} \leq \mu_{\text{baseline}} \quad H_1 : \mu_{\text{custom}} > \mu_{\text{baseline}}$$

(and analogously vs. baseline+RAG)

For cost/efficiency metrics (token usage, inference time):

$$H_0 : \mu_{\text{custom}} \geq \mu_{\text{baseline}} \quad H_1 : \mu_{\text{custom}} < \mu_{\text{baseline}}$$

(and analogously vs. baseline+RAG)

Notation:

- μ_{custom} = population mean for the custom agent
- μ_{baseline} = population mean for the baseline agent
- $\mu_{\text{baseline+RAG}}$ = population mean for baseline with RAG

All tests are one-tailed / one-sided.

Chapter 4

Results

In progress

Chapter 5

Discussion

In progress

Chapter 6

Conlusion

In progress

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